

Urban GIS Portfolio Project 3

Preliminary Spatial Analysis of Land Subsidence Risk in Tehran

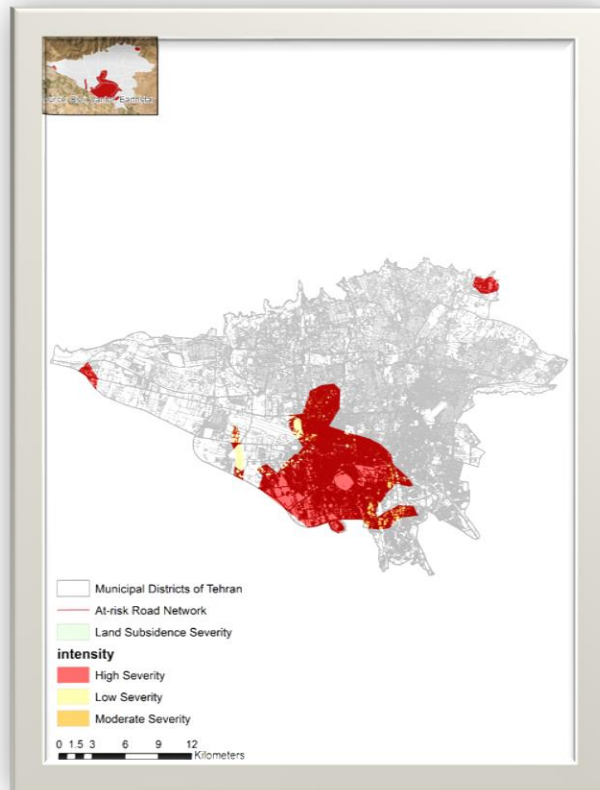
GIS-Based Urban Infrastructure Vulnerability Assessment

Study Area: Tehran, Iran

Status: Research in Progress (Preliminary Results)

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Tools: ArcGIS Pro | Spatial Analysis | Overlay Analysis | Risk Mapping | Cartographic Design



Project Overview

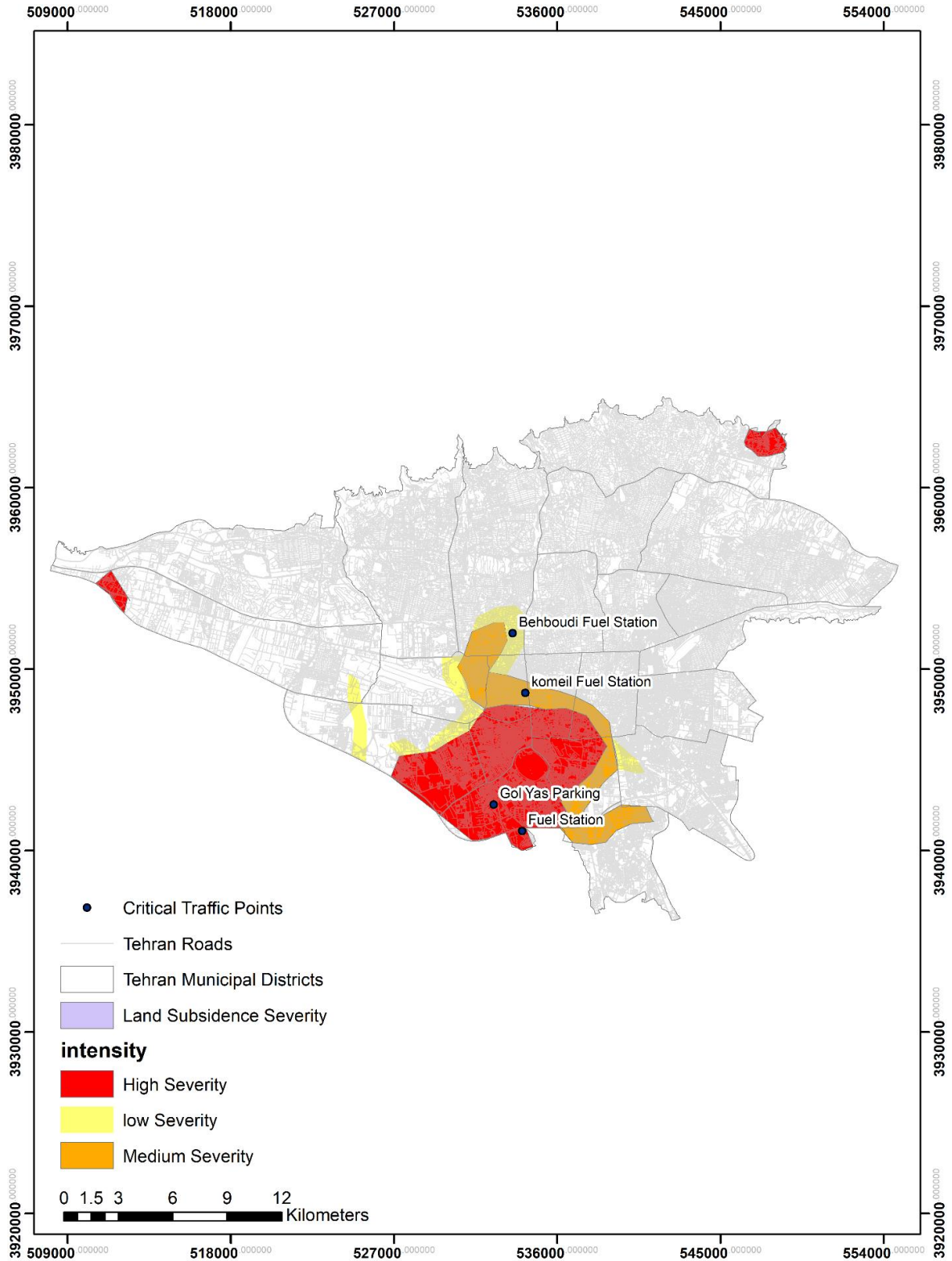
This project presents the preliminary spatial analysis conducted as part of an ongoing research study on land subsidence in Tehran. GIS-based overlay analysis was used to investigate the relationship between land subsidence severity and vulnerable urban infrastructure, including buildings, transportation networks, and critical urban facilities.

Objectives

- Map land subsidence severity across Tehran.
- Assess the exposure of critical urban facilities to subsidence.
- Identify buildings located within high-risk zones.
- Evaluate the vulnerability of the urban road network.

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Spatial Analysis Outputs



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Figure 1. Spatial Distribution of Land Subsidence Severity and Critical Urban Facilities in Tehran

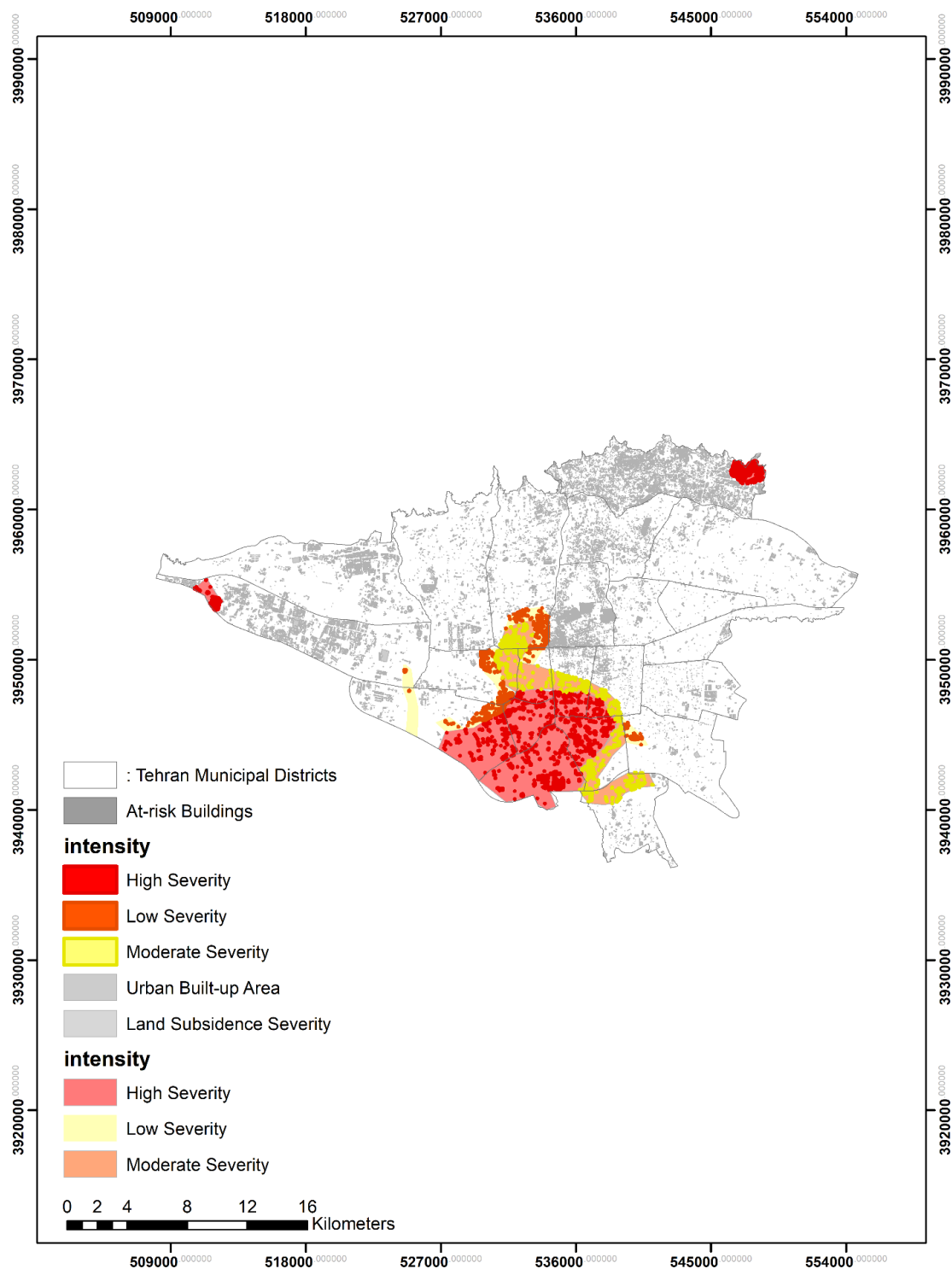


Figure 2. Spatial Assessment of At-Risk Buildings within Land Subsidence Zones in Tehran

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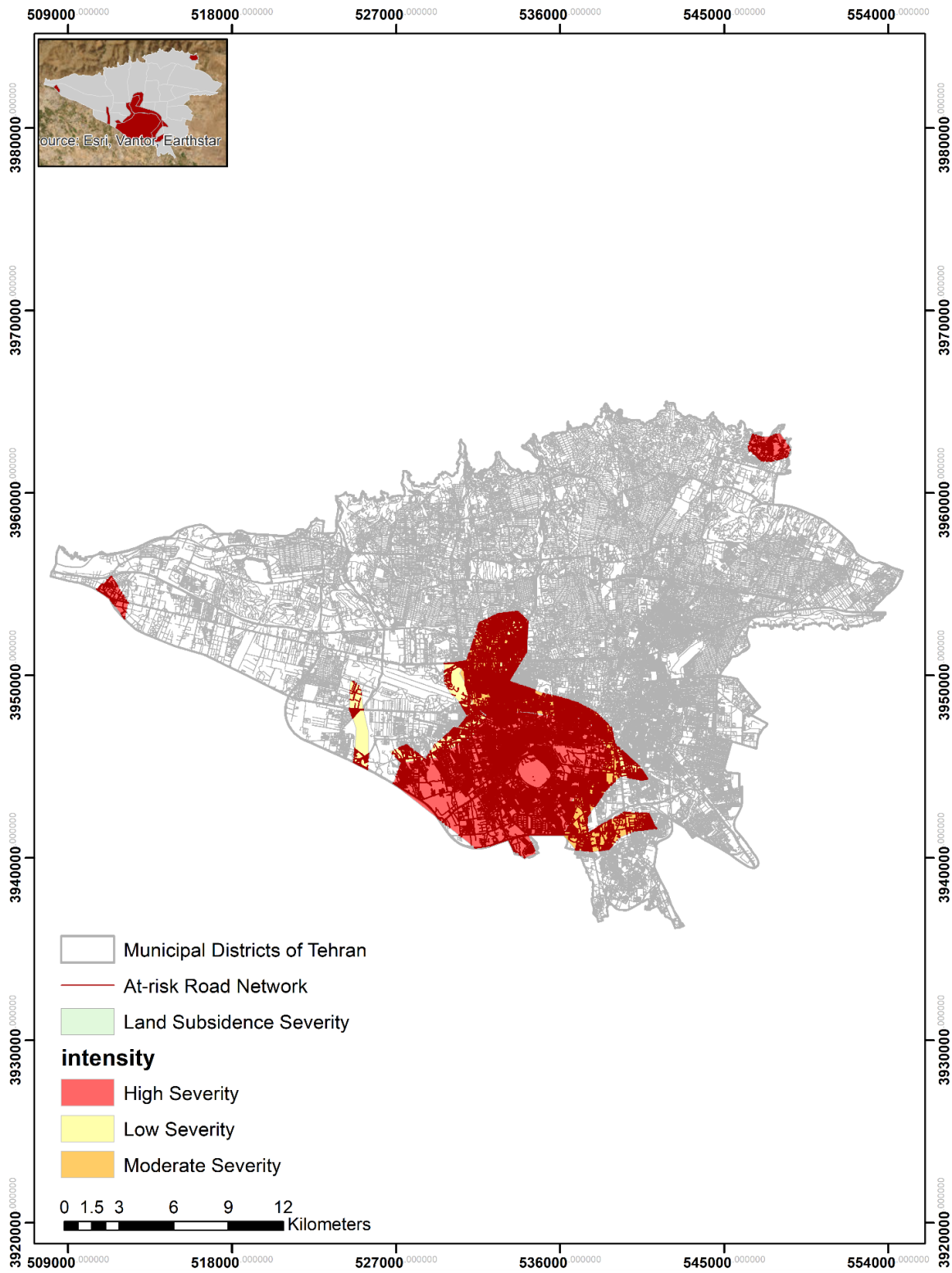


Figure 3. Assessment of Road Network Exposure to Land Subsidence Zones in Tehran

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Project Outcome

The preliminary spatial analysis identified areas where severe land subsidence overlaps with vulnerable urban infrastructure, including buildings, road networks, and critical facilities. The results provide valuable spatial information for future urban risk assessment, infrastructure management, and resilient urban planning.

Key Findings

- High-severity land subsidence zones were successfully identified.
- Several critical urban facilities are located within high-risk areas.
- Vulnerable buildings and road networks were mapped using GIS overlay analysis.
- The generated maps provide a foundation for future land subsidence monitoring and urban risk management.